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Measuring behavioral momentum

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Abstract

The metaphor of behavioral momentum proposes that when ongoing operant behavior is disrupted, changes in response rate are directly related to a force-like aspect of the disruptor and inversely proportional to behavioral mass. Several data sets suggest that differential resistance to change between the components of a multiple schedule satisfies the requirements of a ratio scale and is additive when different disruptors and different dimensions of reinforcement are combined. Differential resistance also provides a basis for scaling force in relation to rate of food presentation between components as a disruptor, and for scaling mass in relation to food rate within a component as a reinforcer. Preference in concurrent chains with terminal links identical to multiple-schedule components also meets the requirements of ratio-scale measurement, is additive when different dimensions of reinforcement are combined, and provides convergent measurement of behavioral mass. © 2002 Published by Elsevier Science B.V.

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1. Introduction

Several times in the course of his long and productive career, E.L. Thorndike remarked that ‘Whatever exists at all exists in some amount, and can be measured’ (Clifford, 1968, p. 283). Following Thorndike’s lead, many researchers have worried about how to measure behavioral attributes, and the topic of measurement has long been prominent in experimental psychology (for a comprehensive and influential treatment, see Stevens, 1951). I have puzzled for almost 30 years over the problem of measuring the resistance to change of ongoing behavior, which I have identified with the traditional construct of response strength. Because efforts to quantify resistance to change have used the metaphor of behavioral momentum as a framework, I begin with a review of the metaphor.

In classical mechanics, the momentum of a moving body is the product of its mass and velocity. In the metaphor of behavioral momentum, asymptotic response rate is taken to be analogous to velocity, and the resistance to change of that rate is related to a behavioral equivalent of mass. The relation between resistance to change and mass is implied by Newton’s second law: $\Delta v = f/m$, where the change in velocity (acceleration or deceleration) of a moving body effected by an external force is inversely proportional to the mass of that body. In the metaphorical parallel, the change of ongoing re-

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sponse rate during disruption by some external variable is directly proportional to the disruptive force and inversely proportional to behavioral mass. Thus, greater resistance to change—i.e. a smaller effect of a given disruptor—implies greater behavioral mass.

Throughout this article, asymptotic baseline response rate is designated B_O, and response rate during disruption is designated B_X. With $\log(B_X/B_O)$ as the measure of Δv, the behavioral equivalent of Newtons' second law is

$$\log(B_X/B_O) = -f/m \quad (1)$$

where the minus sign on the right indicates that the disruptor f decreases (decelerates) response rate. Empirically, the decelerative force-like effect generally increases with increasing values of the disruptor (e.g. rate of response-independent food presented during intercomponent intervals; amount of prefeeding; or number of sessions of extinction). Because the behavior ratio on the left side of Eq. (1) is dimensionless, f and m must be measured in the same units.

Many studies of resistance to change have employed two-component multiple schedules, for several reasons. First, individual subjects are exposed repeatedly to two or more signaled conditions of reinforcement, permitting the experimenter to compare baseline response rates and resistance to change between components within sessions of a single experimental condition rather than between successive conditions that may be separated widely in time. Cohen (1998) has shown that frequent alternating exposure is important for orderly, internally consistent results. Second, the duration and rate of alternation between multiple-schedule components are controlled by the experimenter, whereas in concurrent schedules, the subject can switch freely between alternatives. The absence of experimental control over exposure to the conditions of reinforcement together with unequal times spent at the richer and leaner alternatives may contribute to the diversity of findings on resistance to change in concurrent schedules (for review see Nevin, 1992).

Experiments employing several different species, responses, reinforcers, and a wide variety of disruptors have shown that the effect of a given disruptor is smaller—i.e. resistance to change is greater—in the multiple-schedule component with the greater reinforcer rate or amount (see Nevin, 1979, 1992; Nevin and Grace, 2000a, for review). These between-component comparisons require only ordinal measurement of resistance to change. Interval or ratio scale measurement is needed for quantitative analyses of resistance to change in relation to the magnitude of the disruptor and the conditions of reinforcement in the schedule components, and for evaluation of the terms representing force and mass in the metaphor.

The measure of resistance to change within a schedule component that appears in Eq. (1), $\log(B_X/B_O)$, was used by Nevin et al. (1983) and in all my subsequent work for several reasons. First, it takes baseline response rate into account; second, it avoids floor effects because logarithms range to $-\infty$; and third, functions relating $\log(B_X/B_O)$ to the value of disruptor f (termed resistance functions) are often roughly linear. Nevin (1992) implicitly assumed that $\log(B_X/B_O)$ had the properties of an interval scale when he calculated the slopes of resistance functions and showed that the ratio of their slopes for two components approximated a power function of the reinforcer rate ratio, for average data aggregated across studies.

Slope ratios are not appropriate for analysis if the empirical resistance functions are nonmonotonic, or if the data are insufficient for reliable slope estimation. For example, Grace and Nevin (1997) varied the reinforcer rate ratio over a two-log-unit range across conditions, but used only a single disruptor value in each condition so that only point estimates of slope were available. Moreover, they found that in some resistance tests, response rate increased in one component and decreased in the other so that the slope ratio was negative. Accordingly, they measured relative resistance to change between components as $\log(B_{X1}/B_{O1}) - \log(B_{X2}/B_{O2})$, which I will term differential resistance to change and abbreviate $R_1 - R_2$ for the purposes of this paper.

Grace and Nevin found that $R_1 - R_2$ was a power function of the reinforcer rate ratio for individual pigeons, and argued that $R_1 - R_2$ was preferable to the slope ratio because it could be calculated meaningfully for a wider range of data. Moreover, if $\log(B_X/B_O)$ can be measured on an interval scale, $R_1 - R_2$ may have the properties of a ratio scale, which requires an absolute zero. If the conditions of reinforcement are identical in two multiple-schedule components, $R_1 - R_2$ must equal zero (within measurement error). An absolute zero also permits the operations of addition and multiplication. I now show that $(R_1 - R_2)$ has additive properties when two independent disruptors are combined, and when two separate dimensions of reinforcement are combined.

2. Combining different disruptors

Although many different disruptors have been used in the study of resistance to change, only Nevin and Grace (2000b) and Nevin et al. (2001a) have examined the effects of combining different disruptors. Both studies arranged two-component multiple schedules in one half of each session, with an intercomponent interval (ICI) separating the components. Concurrent chains with terminal links corresponding exactly to the multiple-schedule components were arranged in the other half of each session. This paradigm permits within-subject, within-condition evaluation of preference between the schedule components as well as differential resistance to change. Here I present only the multiple-schedule resistance data; preference will be considered below.

Nevin and Grace (2000b) arranged multiple variable-interval (VI) VI schedules that varied over eight conditions, and Nevin et al. (2001a) arranged multiple variable-ratio (VR) VI schedules that varied over six conditions. Both studies used four pigeons as subjects. After stable baselines were established, response rate was disrupted in three ways: (1) by presenting response-independent food on a variable-time (VT) schedule during the ICI with the baseline schedules remaining in effect; (2) by discontinuing food altogether (Ext); and (3) by presenting VT food during the ICI and also discontinuing food in both components (VT + Ext). The VT and VT + Ext tests occurred in irregular order within each condition of both experiments, but Ext was always third. Baseline performances were reestablished between disruptors.[1]

Fig. 1 shows an example from Nevin and Grace (2000b), condition 1. The upper panel shows average response rates and the lower panel shows

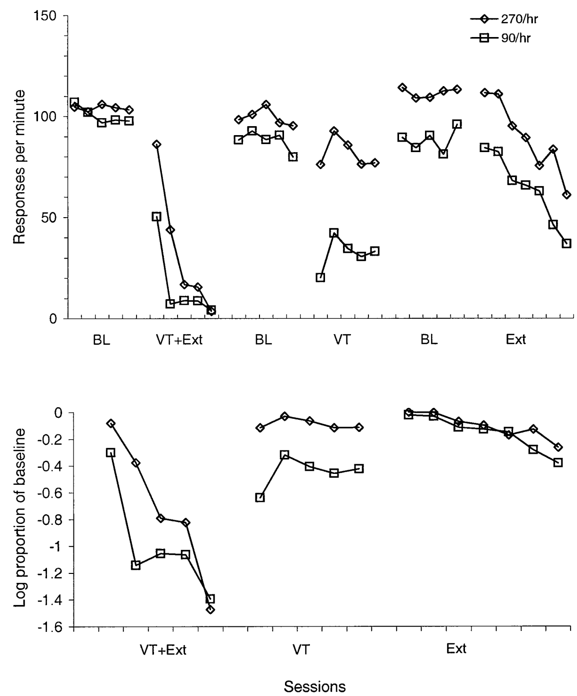

Fig. 1. The upper panel presents average response rates in a two-component multiple VI VI schedule with 270 and 90 reinforcers/h in the components. Data are shown for the last five sessions of baseline training and for the effects of three disruptors: Response-independent food during intercomponent intervals (VT), termination of food reinforcement in the components (Ext), and the combination of VT and Ext. The lower panel presents the data during disruption reexpressed as average log proportions of baseline. Adapted from Nevin and Grace (2000b).

[1] It is important to note that the response-independent food was presented during the ICI and not within components. Cohen et al. (1993) found that although VT reinforcers presented within schedule components disrupted responding, there were no systematic between-component differences in resistance to change. Relatedly, Rescorla and Skucy (1969) showed that after training on a single VI schedule, responding decreased more slowly following a transition to VT than to extinction. I thank Ben Williams for raising this point in discussion at the 2000 SQAB meeting.

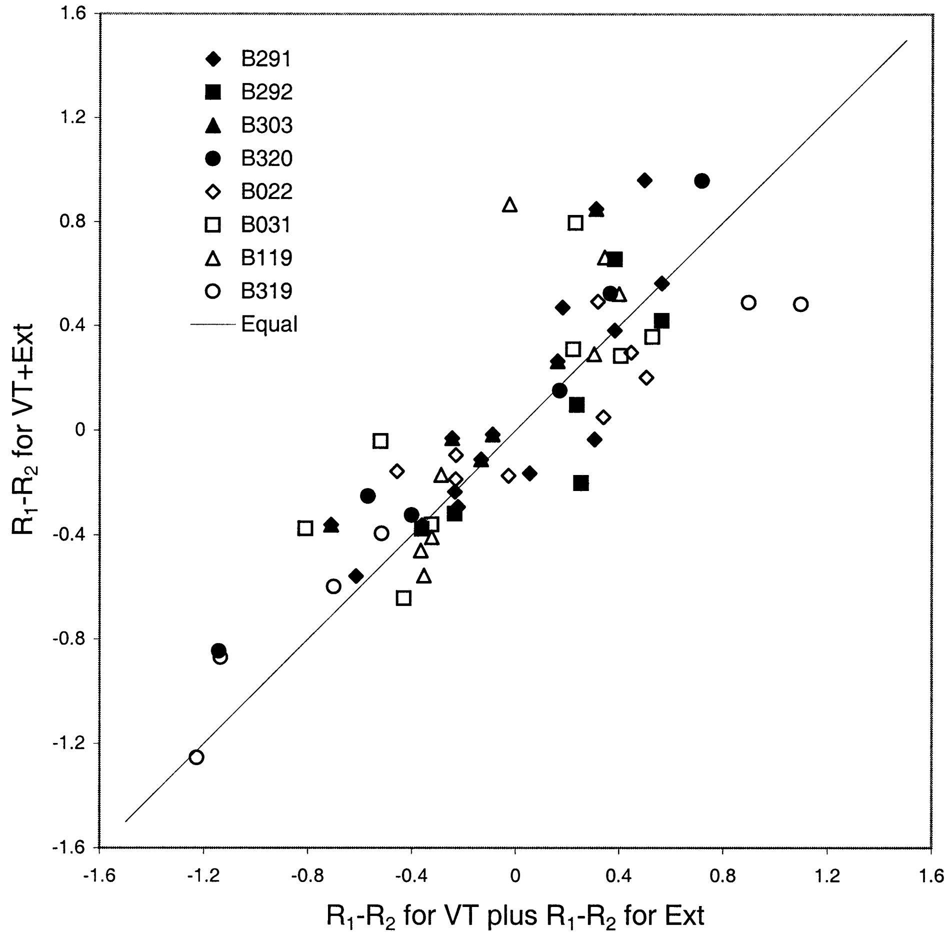

Fig. 2. The sum of between-component differential resistance to VT and to Ext is plotted on the x-axis, and between-component differential resistance to VT + Ext is plotted on the y-axis. The unfilled points are from four pigeons in eight conditions with VI schedules in both components (from Nevin and Grace, 2000b), and the filled points are from four pigeons in six conditions with VR schedules in one component and VI schedules in the other component (from Nevin et al., 2001a,b). The diagonal line indicates exact additivity of differential resistance with respect to the disruptors.

average log proportions of baseline during disruption by VT + Ext, VT, and Ext in that order. Differential resistance, $R_1 - R_2$, is the vertical distance between the pairs of functions in the lower panel. By inspection, behavior is more disrupted in the leaner component than in the richer component, consistent with many previous findings, although the differential effect of extinction is very small in this example. The disruptive effects of VT + Ext are greater than those of VT or Ext alone, and the question is whether the sum of the effects of VT and of Ext would equal the effects of VT + Ext when between-component differential resistance is measured as $R_1 - R_2$.

Fig. 2 displays $R_1 - R_2$ for VT + Ext in relation to the sum of $R_1 - R_2$ for VT and for Ext, based on individual data averaged for the last five sessions of baseline and the first five sessions of disruption. The data points spread out along the line indicating perfect additivity with no obvious systematic deviations in either experiment. Thus, $R_1 - R_2$ is additive, implying ratio-scale measurement (i.e. additivity fails if zero is not defined). In addition, these results suggest that different disruptors combine additively. Appendix A presents the algebra and shows that additivity holds even if the disruptors are effectively different between the two multiple-schedule components.

3. Combining reinforcer dimensions

Do different dimensions of reinforcement also combine additively? The question may be answered by evaluating the separate and combined effects of reinforcer rate and duration, both of which affect resistance to change (see Nevin, 1992, for review). Grace et al. (2002a) arranged three concurrent-chains schedules within each session, where reinforcer duration varied within a condition (signaled by different initial-link key lights on the choice keys). When the keys were red, reinforcer durations were 3.33 and 1.67 s in the left and right terminal links; when the keys were white, reinforcer durations were 2.5 and 2.5 s; and when the keys were green, reinforcer durations were 1.67 and 3.33 s. Thus, differential resistance between left and right keys in red and green components gives the effect of 2:1 versus 1:2 duration ratios. Relative rate of reinforcement on the left and right keys varied between conditions. In conditions 11 and 12, which provide the example here, reinforcer rate ratios were 2:1 and 1:2. Thus, differential resistance within keys between conditions gives the effect of 2:1 versus 1:2 rate ratios.

Resistance to change was evaluated by presenting VT food during the initial links with response-independent presentation of the terminal links, which occurred successively within each condition and hence are equivalent to multiple-schedule components. The average data for conditions 11 and 12 are shown in Fig. 3, with reinforcer rate ratios on the x-axis and with the reinforcer duration ratio as a parameter. Additive combination of reinforcer rate and duration would be demonstrated by parallel lines connecting pairs of data points. The results suggest that the lines are parallel within the standard errors indicated.

Complete individual data for the preceding ten conditions (as well as conditions 11 and 12) are presented by Grace et al. (2002a) in logarithmic generalized matching law form:

$$R_1 - R_2 = a_r \log(r_1/r_2) + a_d \log(d_1/d_2) + \log c \qquad (2)$$

where r and d represent reinforcer rate and duration, subscripted for the two components; a_r is the sensitivity of $R_1 - R_2$ to reinforcer rate; a_d is the sensitivity of $R_1 - R_2$ to reinforcer duration; and $\log c$ represents bias toward one or the other component. To the extent that a_r is the same for the three values of d_1/d_2 (i.e. the functions are parallel as in Fig. 3), the effects of reinforcer rate and amount are additive. There is a good deal of individual variation in the resistance data of Grace et al., but the average values of a_r (slopes) for 2:1 durations (red) and 1:2 durations (green) were essentially identical: 24 versus 24 for conditions with variable-duration components terminating with reinforcement, and 0.39 versus 40 for conditions with constant-duration components. The average slopes for 1:1 durations (white) were somewhat steeper in both sets of conditions, but not consistently so across birds. Thus, it appears that reinforcer rate and duration combine additively to determine $R_1 - R_2$.

4. Scaling force and mass

The conclusions presented above are interdependent. Showing that $R_1 - R_2$ has ratio-scale

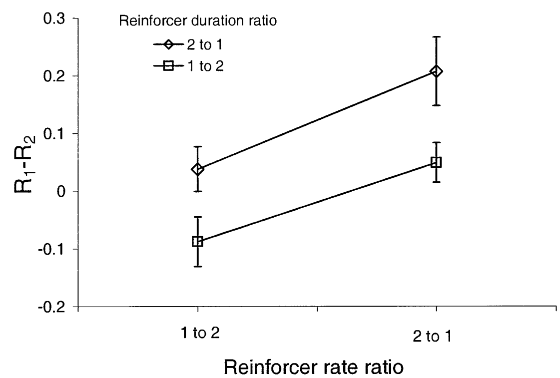

Fig. 3. Effects of the ratio of reinforcer rates on differential resistance to change, with the ratio of reinforcer durations as a parameter. The data are averages of four pigeons from Grace et al. (2002a) (conditions 11 and 12). Parallel lines imply additivity of differential resistance with respect to reinforcer rate and duration.

properties is necessary for the demonstration of additivity, and conversely, the demonstration of additivity validates the measure and its scale properties. Further validation may be achieved by using $R_1 - R_2$ to scale the relation between quantitatively different disruptors such as the rate of VT food during the ICI and force in the momentum metaphor, and at the same time to scale the relation between component reinforcer rate and behavioral mass.

The approach is illustrated with data from two early studies by Nevin (1974) and Nevin et al. (1983). In each study, responding in two-component multiple VI VI schedules was disrupted by presenting response-independent food during intercomponent intervals according to VT schedules. The VT values (in reinforcers/h) varied across successive determinations; baseline response rates were reestablished before each VT test. Both experiments arranged 1-min schedule components, 30-s ICIs, and the same reinforcer duration in the components and in the ICI during tests. Response rates during disruption (B_X) were based on the first hour of session time after introduction of VT food. The average data are presented as log (B_X/B_O) in Fig. 4. By inspection, response rate during disruption is a decreasing function of the rate of VT food in both components, and decreases more relative to baseline in the leaner component at each VT value.

A lot of previous data have suggested that mass is a power function of reinforcer rate (for review see Nevin, 1992; Nevin and Grace, 2000a). Accordingly, I assume that $m_1 = r_1^b$ and $m_2 = r_2^b$, where r designates reinforcer rate subscripted for components 1 and 2. The exponent b characterizes the sensitivity of behavioral mass to reinforcer rate. For dimensional consistency, force must be measured in the same units (to the same power) as mass, so $f = kv^b$ where v designates the rate of VT food presented during the ICI and k scales the relative effectiveness of food as a disruptor and as a reinforcer. Inserting terms into Eq. (1) for each component, $\log(B_{X1}/B_{O1}) = kv^b/r_1^b$ and $\log(B_{X2}/B_{O2}) = kv^b/r_2^b$. Subtracting,

$$R_1 - R_2 = kv^b(r_1^b - r_2^b)/(r_1 r_2)^b \quad (3)$$

Using programmed VI reinforcer rates as r and programmed VT food rates as v, I fitted Eq. (3) to the data shown in Fig. 4 with a nonlinear curve-fitting program (Microsoft Excel Solver). The estimated parameter values are $k = 0.15$, $b = 0.49$, VAC = 0.67. Fig. 5 presents the relation between obtained and predicted values of $R_1 - R_2$, where the filled data points show the result of fitting Eq. (3) to the data of Fig. 4. Although there is a good deal of variation around the line of exact prediction based on these parameter values, it does not appear to be systematic. Thus, to a first approximation, these parameter values serve to scale force and mass in relation to the disruptive effects of VT food during the ICI and the reinforcing effects of VI food in the components of multiple schedules, at least for pigeons.

The force and mass scale parameters appear to apply equally well to multiple concurrent schedules, where each component consists of concurrent VI VI schedules and response rates were disrupted by VT food during the ICI (McLean and Blampied, 1995; McLean et al., 1996). Obtained values of $R_1 - R_2$ were computed from the summed responses for each component, ignoring the distribution of responses across alternatives within a component. Predicted values of $R_1 - R_2$ were derived from Eq. (3) using programmed values of r (summed, like response rates, for the concurrent schedules in each component) and v, and with $k = 0.15$, $b = 0.49$, as for the data of Fig. 4. The results are shown in Fig. 5 as unfilled data points. They are neither more nor less discrepant from prediction than those of the conventional two-component multiple VI VI schedules used to estimate the parameters. These results support findings that resistance to change depends on the total reinforcement obtained in a component (Nevin et al., 1990).

Interestingly, b has been estimated at 0.60 and 0.52 (Grace et al., 2002b, experiments 1 and 2); 0.61 (Nevin and Grace, 1999); and 0.57, 0.52 and 0.50 (Nevin et al., 2001b, experiments 1, 2 and 3) in separate data sets with constant-duration multiple VI VI schedules. These experiments used different disruptors and estimation methods, so

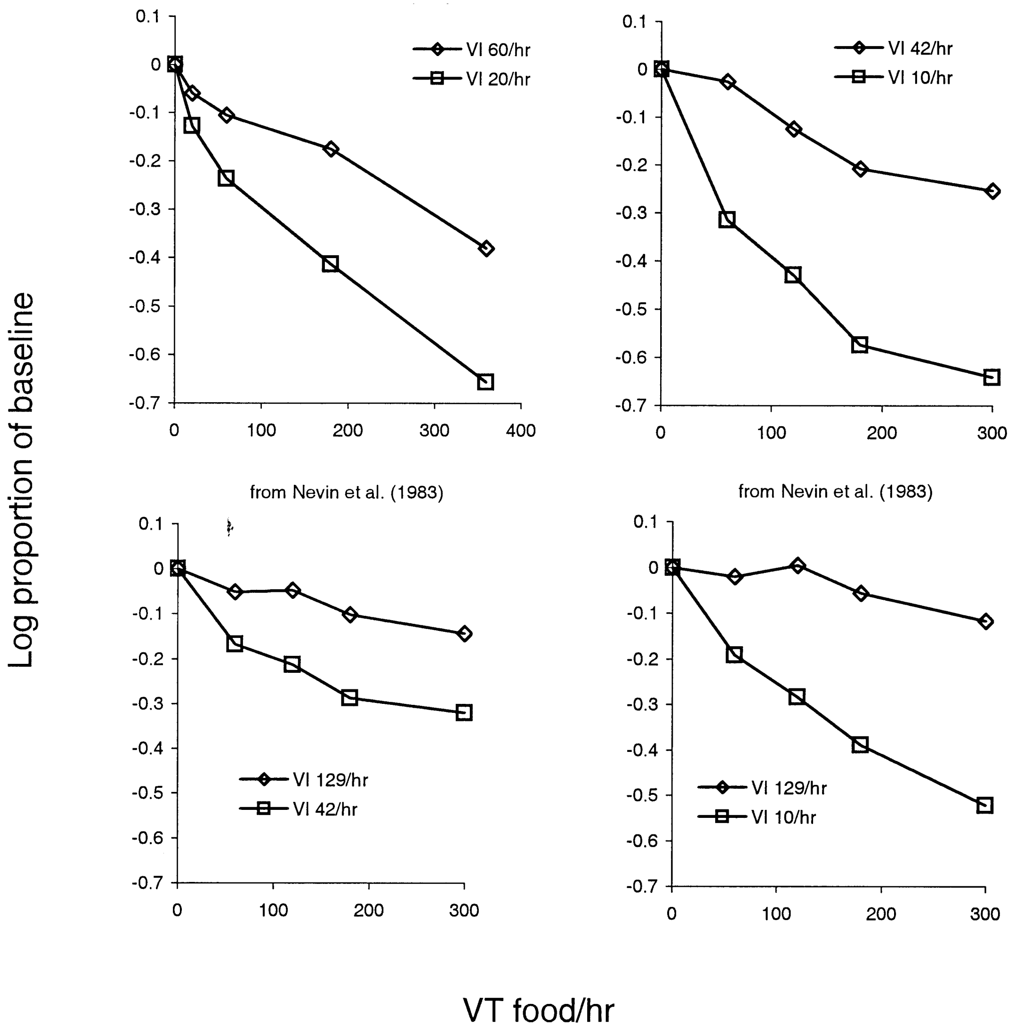

Fig. 4. Response rates in two-component multiple schedules, expressed as average log proportions of baseline, during one-hour probe sessions with response-independent food presented at different rates (VT food/h) during intercomponent intervals. The data are from Nevin (1974) (experiment 1) and Nevin et al. (1983). Component reinforcer rates are indicated in each panel.

there may be some convergence on a generally applicable scale relating behavioral mass to reinforcer rate, at least for pigeons and food reinforcers: to a first approximation, behavioral mass is given by the square root of reinforcer rate.

There is evidence that the reinforcer exponent depends on the experimental paradigm. In a procedure that combined concurrent chains and multiple schedules, Grace and Nevin (2000) found that $R_1 - R_2$ was less sensitive to reinforcer rate ratios when the components were variable in duration and ended with a single reinforcer—in effect, arranging a variable delay from component onset to reinforcement—than when components were constant in duration and included a variable number of reinforcers (see also Grace and Nevin, 1997; Nevin and Grace (2000b); Grace et al., 2002a). Because average reinforcer rate is simply the reciprocal of average delay to reinforcement (i.e. immediacy), the reason for this difference in sensitivity is not clear. Whatever the reason, when multiple-schedule components are arranged as terminal links in concurrent chains, preference between them is also less sensitive to immediacy ratios in variable-duration components than to reinforcer rate ratios in constant-duration components. As will be shown below, the correlation between differential resistance and preference is quite general.

5. Additivity of preference

Before examining relations between differential resistance and preference, I show that combinations of reinforcer rate and duration have additive effects on initial-link preference in concurrent chains. Preference is designated $P_{1,2}$ and is measured as $\log(B_{i1}/B_{i2})$, where B_i designates initial-link responses subscripted according to whether they produce one or another terminal link. Because this measure is zero at indifference, it can meet the requirements of a ratio scale if it also proves to be additive. Grace et al. (2002a) (see description above) measured preference in pre-disruption baselines between components differing in reinforcer duration, with reinforcer rate varied between conditions, before measuring resistance to change. Their preference data were averaged for conditions 11 and 12 exactly as in the $R_1 - R_2$ analysis above and are presented in Fig. 6 in the same format as Fig. 3. As required for additivity, the lines connecting pairs of points are almost exactly parallel, with very small standard errors around the average data of those conditions. Grace et al. found that generalized matching law functions fitted to the individual preference data for the preceding ten conditions were also parallel in nearly all cases, with sensitivity of initial-link choice to terminal-link reinforcer ratios averaging

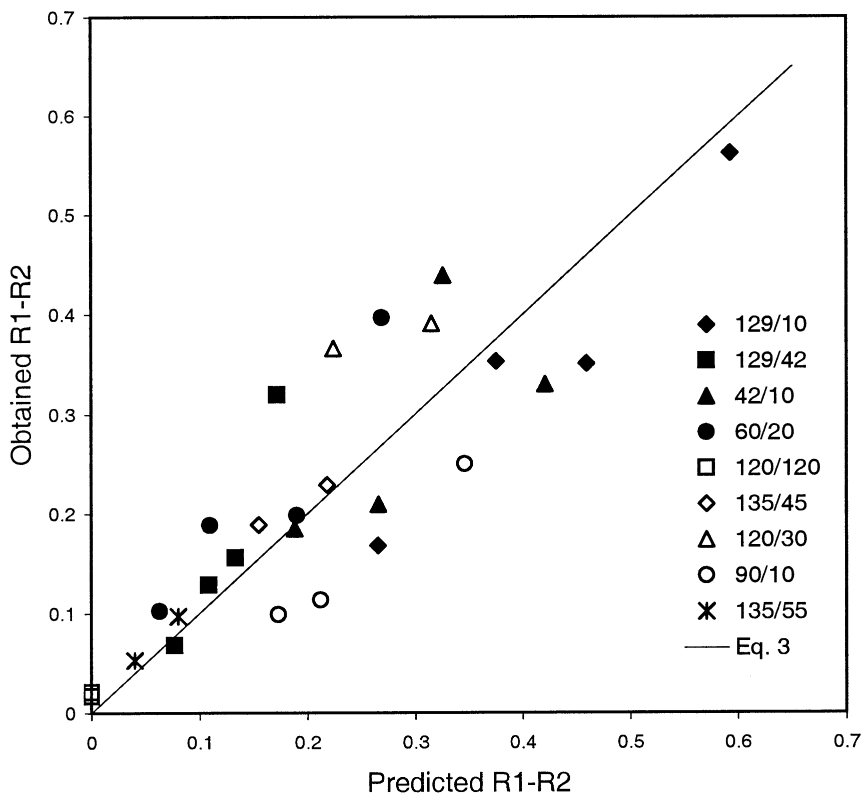

Fig. 5. Values of differential resistance predicted by behavioral momentum theory, with parameters estimated by fitting Eq. (3) to the data of Fig. 4, are displayed on the x-axis, with the corresponding obtained values on the y-axis. Reinforcers/h in the schedule components are given in the legend. Filled data points are based on the data of Fig. 4; unfilled data points are from McLean and Blampied (1995) and McLean et al. (1996).

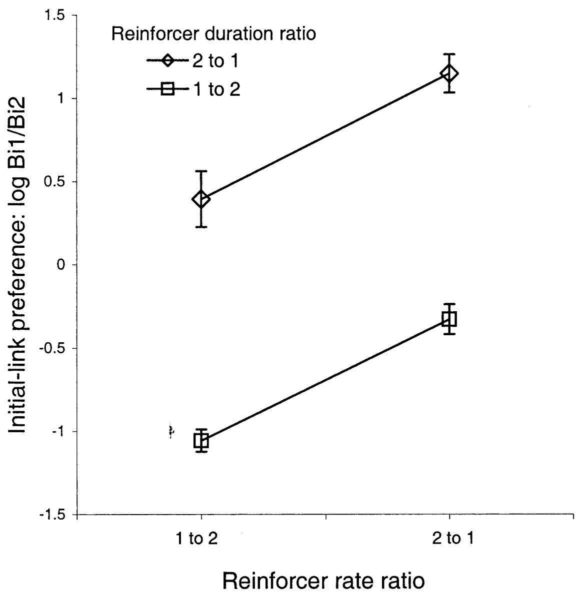

Fig. 6. Effects of the ratio of reinforcer rates on preference in concurrent chains, with the ratio of reinforcer durations as a parameter. The format is the same as Fig. 3. The data are averages of four pigeons from Grace et al. (2002a) (conditions 11 and 12). Parallel lines imply additivity of preference with respect to reinforcer rate and duration.

0.93 and 0.93 for red (2:1) and green (1:2) in the variable-duration conditions, and 1.38 and 1.35 in the constant-duration conditions. As with $R_1 - R_2$, slopes were slightly but not consistently steeper in the equal-duration component. Thus, preference is additive when measured as the log ratio of initial-link response rates, and reinforcer rate and duration combine additively to determine preference as well as differential resistance.

6. Convergent measurement of behavioral mass

Grace and Nevin (1997) suggested that in concurrent chains where the terminal links were identical to the multiple-schedule components arranged for evaluation of resistance, initial-link preference could provide independent, convergent measurement of a central construct corresponding to behavioral mass. (See Nevin and Grace, 2000a, for a metaphorical linkage between inertial mass estimated by resistance to change and gravitational mass estimated by preference.) A number of subsequent studies have obtained independent measures, within subjects and conditions, of resistance to change between components in multiple schedules and preference between those components arranged as terminal links of concurrent chains. Taken as a group, these studies have shown that differential resistance and preference are very highly correlated both within experiments that varied the conditions of reinforcement and between experiments that employed different schedule types or procedural variations (e.g. constant-duration or variable-duration components) that affected both dependent measures. Fig. 7 (from Grace et al., 2002a) presents the data from these studies. Preference $P_{1,2}$ is on the x-axis and differential resistance $R_1 - R_2$ is on the y-axis. Each data point is the average for four pigeons, where preference was determined in five baseline sessions immediately preceding resistance determinations of five sessions with 360 VT food presentations per hr during the ICI or the initial link. The relation can be described as $R_1 - R_2 = 0.29P_{1,2}$. The constant of proportionality is the transform between the ratio scale of preference and the ratio scale of differential resistance to change. Its value has no special significance because $R_1 - R_2$ is directly related to the rate of VT food presentations (see Fig. 4), so that a leaner VT schedule would have yielded a lower value and a richer schedule would have yielded a higher value. The important point is that an inferred construct variously termed response strength, schedule value, or behavioral mass—or more empirically, the effect of a history of reinforcement correlated with a distinctive stimulus—is expressed in two independent ways that are highly correlated and measurable on ratio scales.

The convergence of differential resistance and preference, each of which satisfies criteria for ratio-scale measurement, appears to imply that the inferred behavioral attribute corresponding to mass in the momentum metaphor can also be measured on a ratio scale. However, the absence of experimentally arranged reinforcement does not imply zero mass, as required for ratio scaling. Herrnstein (1970) suggested that activities other than the measured response, such as scratching or defecating, can occur in any situation, and they carry their own reinforcers such as relief of discomfort. Nevin et al. (1990) demonstrated that resistance to change of a designated response depends on all reinforcers in a situation, regardless of their source, and Grimes and Shull (2001) showed that this generalization holds for qualitatively different reinforcers. Therefore, resistance to change may be affected by extraneous or background reinforcers. To take them into account, Eq. (1) may be rewritten as

$$\log B_X/B_O = -f/(m_r + m_o) \quad (4)$$

where the mass term reflects the summed effects of experimentally arranged reinforcers, m_r, and background reinforcers, m_o. If m_o is small relative to m_r, it will have no important consequences for the scaling of force and mass as proposed in Eq. (3). Also, it is reasonable to assume that m_o is the same in multiple-schedule components or terminal links with identical conditions of reinforcement, so its value will not alter the true zero required for ratio scaling of differential resistance and preference. Nevertheless, future research should consider the effects of background reinforcers in efforts to quantify behavioral mass.

7. Fundamental measurement

In a frequently cited treatise on measurement in the physical sciences, Campbell (1920) proposed that an attribute is amenable to fundamental measurement if an empirical operation of combination or concatenation gives results that conform to the rules of arithmetic, and if no other attribute need be measured. These requirements are easily met for the standard physical examples of length and weight. However, density is not additive because two bodies cannot be placed together to yield the sum of their separate densities. Accordingly, density is a derived magnitude given by

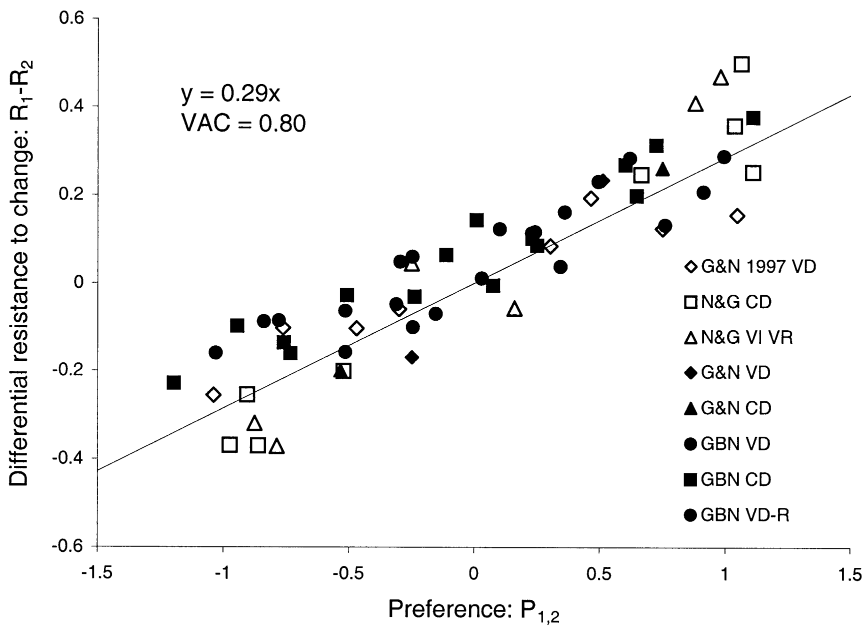

Fig. 7. Differential resistance to change, on the y-axis, in relation to preference, on the x-axis, from all extant data sets where differential resistance and preference were measured within subjects and conditions. Relative reinforcer durations as well as rates varied in GBN. VD designates studies where the terminal links and components varied in duration, ending with a single reinforcer (unfilled data points); VD-R designates a replication. CD designates studies where the terminal links were constant in duration and could include one, many, or no reinforcers (filled data points). VI schedules were used in all cases except NGHM VI VR, which used constant-duration terminal links and components. The fitted regression line summarizes the results. From Grace et al. (2002a).

weight/volume, which depends on fundamental measurement of weight and length.

In his influential overview of measurement issues in psychology, Stevens (1951) suggested that although fundamental measurement scales may be important, they are only instances of ratio scales. In a more formal treatment of measurement, Suppes and Zinnes (1963) stated that the existence of additive operations carries no special weight in measurement theory. Nevertheless, empirical operations of combination that correspond to the numerical operations of addition are rare in behavioral science, so it worth noting that both differential resistance to change and preference meet the requirements of fundamental measurement. As shown above, $R_1 - R_2$ is additive when independent disruptors are combined and when independent dimensions of reinforcement are combined. Moreover, no other attribute need be measured. The only operation required to specify the value of $R_1 - R_2$ is counting responses over equal periods of time, and there is no need to measure time in multiple-schedule components as long as temporal equality is defined by an operation such as the rotation of a cam timer. Likewise, preference measured as $\log(B_{i1}/B_{i2})$ is additive when independent dimensions of reinforcers are combined, and again no other attribute need be measured because initial-link responses B_{i1} and B_{i2} are available concurrently for equal periods of time. It remains to be seen whether fundamental measurement of these terms will be important for future quantification.

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Appendix A

Within-component resistance to change, $\log(B_X/B_O)$, is assumed to be an interval scale measure. Interval scales are unique up to the linear transformation $x' = ax + b$, but additivity fails under linear transformation unless $b = 0$. For example, suppose $x_1 + x_2 = x_3$. But $x'_1 + x'_2 = ax_1 + b + ax_2 + b$ or $a(x_1 + x_2) + 2b$, which is not the same as $x'_3 = ax_3 + b$. If $\log(B_X/B_O)$ is measured on an interval scale, then differential resistance, $R_1 - R_2$, is a ratio scale measure which is unique up to the similarity transformation $x' = ax$. Differential resistance has ratio scale properties because the constant b of the linear transformation is removed by subtraction: If $x'_i = ax_i + b$ and $x'_j = ax_j + b$, then $x'_i - x'_j = a(x_i - x_j)$. Thus, $R_1 - R_2$ can have additive properties.

To show that additivity holds for $R_1 - R_2$ in the context of the momentum model, we take the difference between versions of Eq. (1) written separately for components 1 and 2, with VT food designated force A. For generality, f_A is subscripted for components 1 and 2 to accommodate the possibility that the effective disruptor differs between components, as Nevin et al. (2001b) have argued for extinction:

$$(R_1 - R_2)_A = -f_{A1}/m_1 - (-f_{A2}/m_2) = f_{A2}/m_2 - f_{A1}/m_1 \quad \text{(A1)}$$

Repeating Eq. (A1) for extinction (force B):

$$(R_1 - R_2)_B = f_{B2}/m_2 - f_{B1}/m_1 \quad \text{(A2)}$$

Finally, repeating Eq. (A1) for VT + Extinction (force C):

$$(R_1 - R_2)_C = (f_{A2} + f_{B2})/m_2 - (f_{A1} + f_{B1})/m_1 \quad \text{(A3)}$$

Eqs. (A1) and (A2) sum to Eq. (A3). Therefore, force is additive even when the effective disruptor differs between components. The data of Fig. 2 provide empirical support for the additivity of forces implied by Eq. (A3).

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